

- (ii) Justify the number of significant figures that you have given for your values of k .

.....

 [1]

- (e) It is suggested that the percentage uncertainty in the values of k is 5%.

Using this uncertainty, explain whether your results support the relationship in (d).

.....

 [1]

- (f) It is suggested that

$$k = \frac{hc}{e\lambda}$$

where h is $6.63 \times 10^{-34} \text{ J s}$,
 c is $3.00 \times 10^8 \text{ m s}^{-1}$,
 e is $1.60 \times 10^{-19} \text{ C}$ and
 λ is the wavelength of the light emitted by the LED.

Use your second value of k to determine λ .

$\lambda = \dots\dots\dots \text{ m}$ [1]

